Egg Drops: The Corrected 1–3 versus 2–2 Comparison

A constant-factor analysis for four eggs and N floors

Abstract. This note corrects a subtle but important point in the 1–3 strategy. The first egg is not simply dropped once and then forgotten. We choose a step size S for the first egg, drop it at successive multiples of S, and, when it breaks, the remaining uncertainty is a genuine three-egg egg-drop problem. The three-egg subproblem is therefore allowed to use its optimal adaptive strategy. Comparing this with the stated 2–2 block strategy gives the asymptotic worst-case counts

1–3: 4(2/9)^(1/4) N^(1/4) + O(1) = 2.746356... N^(1/4) + O(1),
2–2: 2√2 N^(1/4) + O(1) = 2.828427... N^(1/4) + O(1).

Thus, within these two strategy families, the corrected 1–3 construction has the smaller leading constant. The advantage is about 2.99% in the leading constant. The difference comes from using the genuinely optimal three-egg subproblem after the first egg breaks, rather than using the paper's simpler uniform-step upper bound for that subproblem.

1. The egg-drop model

There is a critical floor n. Dropping an egg below n leaves it intact, while dropping it at n or above breaks it. An intact egg can be reused; a broken egg is discarded. We minimize the worst-case number of drops. These are the conventions used in the cited talk and paper.

The paper distinguishes dynamic searches, in which later drop locations depend on earlier outcomes, from fixed-step strategies. It also records the standard decision-tree count: with k eggs and d drops, the maximum number of floors that can be resolved is Σ(j=1 to k) C(d,j). We will use that exact dynamic count for the 3-egg and 2-egg subproblems.

2. A useful counting fact

Lemma. With k eggs and d drops, one can distinguish at most—and, in the standard egg-drop model, can achieve—

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M_k(d) = C(d,1) + C(d,2) + ... + C(d,k)
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candidate floors. For k=2 this is d(d+1)/2. For k=3 it is d + d(d−1)/2 + d(d−1)(d−2)/6. Therefore the number of drops required to search m floors satisfies, asymptotically,

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D_2(m) = √(2m) + O(1), D_3(m) = (6m)^(1/3) + O(1).
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The O(1) terms come from the fact that the number of drops is an integer. Only the leading terms matter for the N^(1/4) constants below.

3. The corrected 1–3 strategy

Strategy. Choose a first-egg step size S. Drop the first egg at floors S, 2S, 3S, ... until it breaks. If it breaks at a multiple of S, then the critical floor lies in the preceding block of at most S floors. Since the first egg is now gone, we have three eggs left, and we solve that remaining interval using the optimal three-egg strategy.

This is the correction to the earlier version: the first egg is used repeatedly at an optimized interval S. Once it breaks, the remaining search is not approximated by a linear or fixed-step search; it is the three-egg problem.

In the worst case the first stage costs about N/S drops. The residual interval has at most S−1 candidate floors, so the second stage costs $D_3(S-1)$ drops. Hence

$$T_{13}(N,S) \le \lfloor N/S \rfloor + D_3(S-1).$$

4. Asymptotically optimize S

Using $D_3(S-1) = (6S)^{1/3} + O(1)$, we minimize the continuous approximation

$$f(S) = N/S + 6^{1/3} S^{1/3}.$$

Differentiate:

$$f'(S) = -N/S^2 + (1/3)6^{1/3}S^{-2/3}.$$

Set $f'(S)=0$:

$$N/S^2 = (1/3)6^{1/3}S^{-2/3}.$$

Multiplying by S^2 gives $S^{4/3} = 3N / 6^{1/3}$. Thus

$$S^* = (3 / 6^{1/3})^{3/4} N^{3/4} = 1.456475... N^{3/4}.$$

The second derivative is positive at this point, so this is the minimum of the continuous approximation. Substituting S* back into f gives

$$f(S^*) = 4(6^{1/3}/3)^{3/4} N^{1/4}$$

which simplifies to

$$T_{13}(N) = 4(2/9)^{1/4} N^{1/4} + O(1) = 2.7463561919... N^{1/4} + O(1).$$

5. The 2–2 block strategy

Now divide the N floors into blocks of F consecutive floors. Treat each block as one superfloor. There are $M = \lfloor N/F \rfloor$ superfloors. First solve the two-egg problem on the M superfloors. Once the critical superfloor is identified, solve the two-egg problem inside that block of at most F floors.

Using the optimal two-egg subroutine, the worst-case count is

$$T_{22}(N,F) = D_2(\lfloor N/F \rfloor) + D_2(F).$$

Since $D_2(m)=\sqrt{(2m)}+O(1)$,

$$T_{22}(N,F) = \sqrt{2}\,\sqrt{(N/F)} + \sqrt{2}\,\sqrt{F} + O(1).$$

6. Optimize F

Ignore the O(1) terms and minimize

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g(F) = √2 (N^(1/2)F^(−1/2) + F^(1/2)).
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Differentiating gives
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g′(F) = (√2/2)(−N^(1/2)F^(−3/2) + F^(−1/2)).
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Setting this equal to zero gives F=N^(1/2). Thus
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T_22(N) = 2√2 N^(1/4) + O(1) = 2.8284271247... N^(1/4) + O(1).
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7. Compare the constants

Strategy	Leading asymptotic count	Constant
1–3	4(2/9)^(1/4) N^(1/4)	2.746356...
2–2	2√2 N^(1/4)	2.828427...

The ratio of the leading constants is

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2.746356... / 2.828427... = 0.970983...
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So the 1–3 construction uses about 97.10% as many leading-order drops as the 2–2 construction, a reduction of about 2.99% in the leading constant.

Conclusion: Under the stated strategy definitions, the corrected 1–3 approach has the smaller asymptotic constant.

8. Why the earlier calculation could go wrong

There are two different levels of approximation in the cited paper. The paper proves, for its uniform fixed-step recursive strategy, the upper bound P_1(k) ≤ ■k N^(1/k)■. For k=3 this gives a 3-egg subproblem bounded by roughly 3S^(1/3), rather than the sharper (6S)^(1/3) coming from the exact dynamic decision-tree count. The paper explicitly notes that its fixed-step methods are distinct from the earlier dynamic search.

If one uses only that coarser fixed-step bound inside the 1–3 calculation, one obtains a different leading constant. That is not the corrected 1–3 comparison asked for here, because the strategy says that after the first egg breaks we solve the three-egg problem. The natural interpretation is therefore to use the optimal three-egg subproblem.

Likewise, for the 2–2 construction the talk emphasizes the dynamic triangular-number two-egg strategy: the step lengths decrease by one after each survival. This gives D_2(F)=√(2F)+O(1), rather than the coarser fixed-step estimate F^(1/2) with an extra factor.

9. A numerical check

N = 10^8	Continuous optimum	Nearby integer choice	Drops
1–3	S* ≈ 1,456,475	S = 1,449,276	275
2–2	F* = 10,000	F = 9,448	282

For this example, the exact integer subproblem counts near the asymptotic choices give 275 drops for the 1–3 construction and 282 drops for the 2–2 construction. This is only an illustration; the asymptotic comparison is the main result.

10. Final perspective

Both constructions have order N^(1/4) because the first stage and second stage can be balanced. For 1–3, the first stage costs about N/S while the optimal three-egg search costs about S^(1/3), so balancing forces S to be of order N^(3/4). For 2–2, the two stages cost about (N/F)^(1/2) and F^(1/2), so balancing forces F to be of order N^(1/2). The exponents agree, and the contest is decided entirely by the constants.

The important conceptual correction is therefore: **1–3 means an optimized first-egg jump search followed, after the first egg breaks, by a genuine three-egg optimal search.** It does not mean that the first egg is dropped only once.

References

1. Steven J. Miller, “Egg Drop Mathematics: It IS all its cracked up to be!”, Hampshire College talk, 28 July 2026.

2. Xiangwen Cao, Zongyun Chen, and Steven J. Miller, “Egg Drop Problems: They Are All They Are Cracked Up to Be!”, Fibonacci Quarterly, compiled May 10, 2026.

The source material distinguishes dynamic search from fixed-step strategies and gives the decision-tree count Σ C(d,j) for k eggs and d drops; the present note specializes those facts to the requested 1–3 versus 2–2 comparison.